

Number System – Detailed Notes

Computer Science Grade 9

Introduction

Numbers are used everywhere in our daily life. We use numbers to count objects, calculate prices, measure distances, tell time, and perform scientific calculations. In ancient times, people used fingers, stones, sticks, and knots of rope for counting. Later, different number systems were developed.

A **number system** is a method of representing numbers using a set of symbols and rules.

Different number systems are used in computers and digital devices. Computers, smartphones, tablets, calculators, and other digital systems mainly use the **binary number system**.

Important Terms

Digit

A **digit** is a single symbol used in a number system.

Example:

- Decimal system digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Binary system digits: 0 and 1

Computer Word

A **computer word** is a group of bits processed together by a computer.

Base or Radix

The **base** or **radix** of a number system is the total number of digits or symbols used in that number system.

Example:

- Decimal system → base 10
- Binary system → base 2
- Octal system → base 8
- Hexadecimal system → base 16

Concept of Number System

A number system is a writing system used to express numbers with the help of symbols and rules.

Different number systems use different digits and different bases.

Example:

- Decimal system uses 10 digits.
- Binary system uses 2 digits.
- Octal system uses 8 digits.
- Hexadecimal system uses 16 symbols.

The decimal number system is commonly used by humans because it is easy to understand. Computers use binary because electronic devices work with two states:

- ON (1)
- OFF (0)

Types of Number System

There are mainly four types of number systems:

1. Decimal Number System
2. Binary Number System
3. Octal Number System
4. Hexadecimal Number System

1. Decimal Number System

The decimal number system is the most commonly used number system in daily life.

Features

- Base = 10
- Uses digits from 0 to 9
- Each position has a place value based on powers of 10

Place Values in Decimal System

Position	Place Value
Units	10^0
Tens	10^1
Hundreds	10^2
Thousands	10^3

Example

$$\begin{aligned} & (719)_{10} \\ &= (7 \times 10^2) + (1 \times 10^1) + (9 \times 10^0) \\ &= (7 \times 100) + (1 \times 10) + (9 \times 1) \\ &= 700 + 10 + 9 \\ &= 719 \end{aligned}$$

2. Binary Number System

The binary number system is used by computers.

Features

- Base = 2
- Uses only two digits:
 - 0
 - 1
- Each position represents powers of 2.

Example

$$(110111)_2$$

Why Computers Use Binary

Electronic circuits have only two states:

- ON
- OFF

Therefore, binary numbers are suitable for computers.

3. Octal Number System

The octal number system is a base-8 number system.

Features

- Base = 8
- Uses digits from 0 to 7

Examples

- $(207)_8$
- $(5706)_8$
- $(601022)_8$

4. Hexadecimal Number System

The hexadecimal number system is a base-16 number system.

Features

- Base = 16
- Uses 16 symbols:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Hexadecimal Values

Hexadecimal	Decimal
A	10
B	11
C	12
D	13
E	14
F	15

Examples

- $(8F5)_{16}$
- $(AC4)_{16}$
- $(B52)_{16}$
- $(ACD)_{16}$

Number System Chart

Number System	Base	Symbols/Digits Used
Binary	2	0, 1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal	16	0–9 and A–F

Application of Number System Conversion

Number system conversion is very important in computer science and digital technology.

Uses of Number System Conversion

1. Computers

Computers process data in binary form.

2. Digital Electronics

Digital circuits use binary and hexadecimal values.

3. Networking

IP addresses are represented using decimal and hexadecimal forms.

4. Cryptography

Used for secure data transmission.

5. Graphics and Multimedia

Used in image processing and color representation.

6. Signal Processing

Used in communication systems.

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Binary Calculation

There are four main types of binary calculations:

1. Binary Addition
2. Binary Subtraction
3. Binary Multiplication
4. Binary Division

Binary Addition

Binary addition is similar to decimal addition.

Rules of Binary Addition

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	10

Note:

$1 + 1 = 10$ means:

- Write 0
- Carry 1

Steps of Binary Addition

Step 1

Arrange the binary numbers properly.

Step 2

Start adding from the rightmost digit.

Step 3

Follow binary addition rules.

Step 4

Carry 1 whenever necessary.

Step 5

Continue until all digits are added.

Example of Binary Addition

Add:

$$1100 + 0101$$

Solution:

$$\begin{array}{r} 1100 \\ + 0101 \\ \hline 10001 \end{array}$$

Therefore,

$$(1100)_2 + (0101)_2 = (10001)_2$$

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Binary Subtraction

Binary subtraction is similar to decimal subtraction.

Rules of Binary Subtraction

A	B	A – B
0	0	0
1	0	1
1	1	0
0	1	1 (borrow 1)

Steps of Binary Subtraction

Step 1

Arrange numbers properly.

Step 2

Start subtracting from the rightmost digit.

Step 3

Borrow 1 from the left when necessary.

Step 4

Continue until subtraction is complete.

Example of Binary Subtraction

Subtract:

$$(10111)_2 - (101)_2$$

$$\begin{array}{r} 10111 \\ - 00101 \\ \hline 10010 \end{array}$$

Therefore,

$$(10111)_2 - (101)_2 = (10010)_2$$

Binary Multiplication

Binary multiplication is similar to decimal multiplication, but it uses only 0 and 1.

Rules of Binary Multiplication

A	B	A × B
0	0	0
0	1	0
1	0	0
1	1	1

Steps of Binary Multiplication

Step 1

Write the numbers properly.

Step 2

Multiply each digit starting from the right side.

Step 3

Add all partial products.

Example of Binary Multiplication

Multiply:

$$(101)_2 \times (11)_2$$

$$\begin{array}{r} 101 \\ \times 11 \\ \hline 101 \\ + 101 \\ \hline 1111 \end{array}$$

Therefore,

$$(101)_2 \times (11)_2 = (1111)_2$$

Binary Division

Binary division is similar to decimal division.

Rules of Binary Division

A	B	$A \div B$
0	1	0
1	1	1

Division by 0 is not possible.

Example of Binary Division

Divide:

$$(1010)_2 \div (10)_2$$

$$101 \div 10 = 10$$

Therefore,

$$(1010)_2 \div (10)_2 = (101)_2$$

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Number System Conversion

Number system conversion means changing a number from one number system to another.

The four major number systems are:

- Binary (Base 2)
- Octal (Base 8)
- Decimal (Base 10)
- Hexadecimal (Base 16)

Conversions can be done between all number systems.

1. Decimal to Binary Conversion ($D \rightarrow B$)

Method

1. Divide the decimal number by 2.
2. Write the remainder.
3. Continue dividing the quotient by 2.
4. Repeat until quotient becomes 0.
5. Write the remainders from bottom to top.

Example

Convert $(46)_{10}$ into binary.

2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
2	1	1
	0	

Reading remainder from bottom to top:

101110

Therefore,

$$(46)_{10} = (101110)_2$$

2. Decimal to Octal Conversion ($D \rightarrow O$)

Method

1. Divide the decimal number by 8.
2. Write the remainder.
3. Continue division until quotient becomes 0.
4. Write remainder from bottom to top.

Example

Convert $(345)_{10}$ into octal.

8	345	1
8	43	3
8	5	5
	0	

Reading remainder from bottom to top:

531

Therefore,

$$(345)_{10} = (531)_8$$

3. Decimal to Hexadecimal Conversion ($D \rightarrow H$)

Method

1. Divide the decimal number by 16.
2. Write the remainder.
3. Continue division until quotient becomes 0.
4. Write remainder from bottom to top.
5. If remainder is 10–15, replace with A–F.

Hexadecimal Values

Decimal	Hexadecimal
10	A
11	B
12	C
13	D
14	E
15	F

Example

Convert $(88)_{10}$ into hexadecimal.

16	88	8
16	5	5
	0	

Reading remainder from bottom to top:

58

Therefore,

$$(88)_{10} = (58)_{16}$$

4. Binary to Decimal Conversion ($B \rightarrow D$)

Method

1. Multiply each binary digit by powers of 2.
2. Add all the products.

Example

Convert $(10011)_2$ into decimal.

Binary Digit	Power of 2	Product
1	$2^4 = 16$	16
0	$2^3 = 8$	0
0	$2^2 = 4$	0
1	$2^1 = 2$	2
1	$2^0 = 1$	1

Total = $16 + 0 + 0 + 2 + 1 = 19$

Therefore,

$(10011)_2 = (19)_{10}$

5. Binary to Octal Conversion ($B \rightarrow O$)

Method

1. Group binary digits into groups of 3 from right to left.
2. Convert each group into octal.

Binary to Octal Table

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Example

Convert $(110100)_2$ into octal.

Group into 3 digits:

110 100

Binary	Octal
110	6
100	4

Therefore,

$$(110100)_2 = (64)_8$$

6. Binary to Hexadecimal Conversion ($B \rightarrow H$)

Method

1. Group binary digits into groups of 4 from right to left.
2. Convert each group into hexadecimal.

Binary to Hexadecimal Table

Binary	Hexadecimal
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Example

Convert $(100000110101)_2$ into hexadecimal.

Group into 4 digits:

1000 0011 0101

Binary	Hexadecimal
1000	8
0011	3
0101	5

Therefore,

$$(100000110101)_2 = (835)_{16}$$

7. Hexadecimal to Decimal Conversion ($H \rightarrow D$)

Method

1. Convert hexadecimal symbols into decimal values.
2. Multiply each digit by powers of 16.
3. Add all the products.

Example

Convert $(1AC)_{16}$ into decimal.

$A = 10$

$C = 12$

Hex Digit	Power of 16	Product
1	$16^2 = 256$	256
$A = 10$	$16^1 = 16$	160
$C = 12$	$16^0 = 1$	12

Total = $256 + 160 + 12 = 428$

Therefore,

$(1AC)_{16} = (428)_{10}$

8. Hexadecimal to Binary Conversion ($H \rightarrow B$)

Method

1. Convert each hexadecimal digit into 4-bit binary.
2. Combine all binary groups.

Example

Convert $(9A3)_{16}$ into binary.

Hexadecimal	Binary
9	1001
A	1010
3	0011

Combining:

100110100011

Therefore,

$$(9A3)_{16} = (100110100011)_2$$

9. Hexadecimal to Octal Conversion ($H \rightarrow O$)

Method

1. Convert hexadecimal to binary.
2. Convert binary to octal.

Example

Convert $(2F)_{16}$ into octal.

Step 1: Convert hexadecimal to binary

Hexadecimal	Binary
2	0010
F	1111

Combined binary:

00101111

Step 2: Group into 3 bits

000 101 111

Binary	Octal
000	0
101	5
111	7

Therefore,

$$(2F)_{16} = (57)_8$$

10. Octal to Binary Conversion ($O \rightarrow B$)

Method

1. Convert each octal digit into 3-bit binary.
2. Combine all binary groups.

Example

Convert $(64)_8$ into binary.

Octal	Binary
6	110
4	100

Combining:

110100

Therefore,

$$(64)_8 = (110100)_2$$

11. Octal to Hexadecimal Conversion ($O \rightarrow H$)

Method

1. Convert octal to binary.
2. Convert binary to hexadecimal.

Example

Convert $(57)_8$ into hexadecimal.

Step 1: Convert octal to binary

Octal	Binary
5	101
7	111

Combined binary:

101111

Step 2: Group into 4 bits

0010 1111

Binary	Hexadecimal
0010	2
1111	F

Therefore,

$$(57)_8 = (2F)_{16}$$

12. Octal to Decimal Conversion ($O \rightarrow D$)

Method

1. Multiply each octal digit by powers of 8.
2. Add all the products.

Example

Convert $(157)_8$ into decimal.

Octal Digit	Power of 8	Product
1	$8^2 = 64$	64
5	$8^1 = 8$	40
7	$8^0 = 1$	7

Total = $64 + 40 + 7 = 111$

Therefore,

$(157)_8 = (111)_{10}$

Shortcut Conversion Table

From	To	Method
Decimal	Binary	Divide by 2
Decimal	Octal	Divide by 8
Decimal	Hexadecimal	Divide by 16
Binary	Decimal	Powers of 2
Octal	Decimal	Powers of 8
Hexadecimal	Decimal	Powers of 16
Binary	Octal	Group 3 bits
Octal	Binary	3-bit binary
Binary	Hexadecimal	Group 4 bits
Hexadecimal	Binary	4-bit binary
Octal	Hexadecimal	Through binary
Hexadecimal	Octal	Through binary